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Can Market Power Really be Estimated?

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Abstract. In a previous paper in this *Review*, Hyde and Perloff ask the question, "Can Market Power be Estimated?" using the "structural model" or two proposed alternatives. They find that none of the three methods produce consistent, meaningful results in the food and beverage industries and that simulations suggest great sensitivity of results to model misspecification. This paper offers an explanation for these disappointing results. The difficulty is that the structural model (and its proposed alternatives) is based on an overly-ambitious estimating form that over-simplifies the diversity and true complexity of oligopoly pricing. By neglecting accounting data on costs and by arbitrarily relying on an assumption that outputs are set by subtracting a constant proportion of the difference between Price and Marginal Revenue, the "structural model" and proposed alternatives try to fit an elegant form to a messy world with predictably disappointing results.

Key words: Market power measurement, oligopoly pricing.

JEL#: L11, L12, L13

According to the mainstream tradition in industrial organization, oligopoly behavior is too varied and complex to be understood with a single simple maximizing model. Optimal pricing and behavior depends on rivals' anticipated responses, which being subjective, are inherently unobservable. Thus the field was founded and developed with the much more modest goals of cataloging and measuring different aspects of industry structures and correlating these with outcomes that appear to be related to the exercise of monopoly power (Mason, 1957). The fundamental theorem of the field for many years was the very limited idea that greater concentration in an industry made it more likely that market power was being exercised there (Bain, 1956). The lack of ambition in the field was not a character flaw or lack of imagination, but a response to the perceived inability to write down an estimatable structural model that would capture the diversity and complexity of oligopoly behavior.

This caution and modesty of reach has been challenged with a new paradigm that promises to reduce the complexity of the oligopoly problem by estimating a general model of industry pricing within which it is possible to model competition, monopoly, and all degrees of oligopoly in between (Bresnahan, 1982; Lau, 1982). In so doing, the field of Industrial Organization has been given an empirical framework with which to eliminate the apparent indeterminacy of the oligopoly result. This so called "structural model" has won wide acceptance (Bresnahan, 1989).

The "structural model" is in fact doubly ambitious since it tries to understand oligopoly behavior in a model in which marginal cost is assumed to be unmeasurable. The difficulties in measuring marginal cost using accounting data are well known: the most prominent problems are that accounting data do not use the same cost concepts as those of the economist and they are incapable of tracing short run marginal cost in its upward sloping region, which is the economically interesting part of the curve. Nonetheless, the tradition in the field of industrial organization held that average variable cost measured from accounting data was an adequate proxy for the inherently unmeasurable marginal cost. (See Domowitz *et al.*, 1986) for a recent example.) The ambition of the structural model is to simultaneously estimate, without use of accounting data, three factors affecting pricing: 1) a marginal cost curve for a typical firm; 2) a demand curve for the industry; and 3) a parameter describing oligopoly behavior.

Hyde and Perloff in their recent paper in this Review, note that the structural model in its application has certain drawbacks, most notably, that it is sensitive to slight variations in its specification, is "data hungry," and does not produce meaningful results in many circumstances. They thus correctly ask the question, "Can Market Power be Estimated?" The apparent answer that they give is, "no." In their simulations, the structural model is quite sensitive to model misspecification. In their attempts to apply the structural model to four digit industries in the food and beverage sector, they are unable to derive meaningful results. The proposed alternatives for the structural model that use less data they find similarly incapable of producing meaningful results in many circumstances.

This paper offers a plausible explanation for the results that Hyde and Perloff published namely – that the structural model (and, by extension, its proposed alternatives) is not nearly as general as is often assumed. It does not capture the diversity and complexity of oligopoly pricing and is overly-ambitious in its attempt to discover market power in an industry while using a minimum of data.

I. The Structural Model Described

More than perhaps any other field, economics is willing to make inferences about the observable world by imposing a rigid and generally unquestioned theoretical structure on data. This practice is generally justified by "as if" sorts of arguments and the axiomatic belief that all behavior is maximizing (Machlup, 1966.) By using basic economic theory, papers in economics journals are able to find patterns and to tease out of data, relationships that would otherwise be invisible.

This is the practice used by the so-called "structural model." Oligopoly pricing is assumed to be the result of equilibrium firm level profit maximization. The model is illustrated in Figure 1, based on the putatively unobservable total cost function of firms in an industry and the industry total revenue function. The slope of the total revenue function is, definitially, marginal revenue: $P + (\partial P/\partial Q)Q$. Its value is the slope of line BB. The slope of the ray from the origin, AA, is product price,

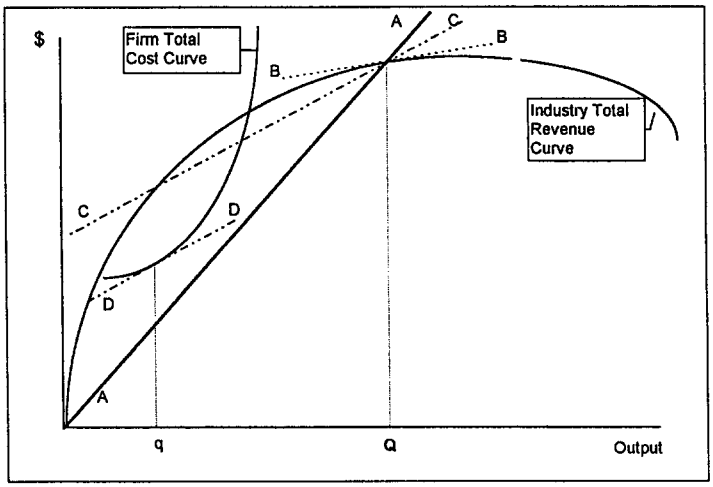

Fig. 1. **The method of the structural model.**
Total revenue at industry output, Q, defines line AA, whose slope is the product price, P, presumed identical for all sellers. At Q, the industry total revenue curve is tangent to BB, whose slope is $P + Q\partial P/\partial Q$. Line CC has a slope $P + \lambda Q\partial P/\partial Q$. The typical firm is presumed to make output decisions by finding that output, q, at which the slope of the slope of its total cost curve, DD, is the same as that of line CC.

P, presumed equal for all firms in the industry. The difference between the two slopes is, then, $(\partial P/\partial Q)Q$. In the structural model, equilibrium behavior requires that a firm compare its marginal cost (the slope of line DD) with an appropriate measure of marginal revenue. If q is consistent with line DD being parallel to AA, the industry is characterized as competitive. If the parallelism occurs between DD and BB, the industry is presumed to be a collusive oligopoly. In the diagram shown, DD is parallel to a line CC whose slope is between that of AA and BB. Thus the industry would be characterized as having market power, but less than complete monopoly power.[1] In mathematical terms, equilibrium is described by:

$$MC_i = P + \lambda(\partial P/\partial Q)Q \qquad \forall i$$

where i indexes the firms in the industry.

The structural model is general in that the unknown λ can range from 0 (representing competitive behavior) to 1 (representing a monopoly result.) If the shapes of the total cost and revenue curves were known, it would be a simple matter to compare the slopes of AA, BB, and DD, all of which would be well defined, and thereby determine the degree to which DD approached AA or BB. Since the total cost and revenue curves are not known, the structural model requires that they be

[1] The reader might note that in the diagram shown, it appears that the typical firm is making a loss at equilibrium. However, since the height of the total cost curve is presumed unknowable, this inference should not be made.)

estimated simultaneously with λ, the proportion of the distance that the slope of CC lies between that of AA and BB. However, the only reliable pieces of information assumed available to the observer are presumed to be industry output, Q, and industry price, the slope of AA.

A single price and quantity observation is not sufficient to estimate the slopes of the lines of interest. As would be true in any econometric study, in order for the researcher to observe the shapes of the total cost and total revenue curves (and through them, λ) price and quantity must not be constant. The structural model assumes that different data points are generated by a combination of the following two forces: 1) A change in input prices alters the total cost curve by changing the minimum cost input combination and total payments made to the factors of production; or 2) a change in demand shifts the industry total revenue curve, thus changing the level of marginal revenue. Either can push the industry out of equilibrium because in either cases, the parallelism between CC and DD is broken. By imposing the requirement that the slope of CC always be the same proportion of the difference between the slopes of AA and BB and by assuming that all observations are produced by a parallelism between DD and CC (that is, all firms are individually maximizing profits subject to the proportionality constraint) repeated shifts of the factors underlying the total cost curve and the total revenue curve will reveal the shapes of the two functions along with the critical proportion that produces CC from AA and BB.

Not surprisingly, the first problem that the structural model deals with is that of identification. Only some ways that the total cost and total revenues can be shifted by outside factors are capable of revealing the shapes of the curves and the critical proportion (Bresnahan, 1982; Lau, 1982.) For example, if exogenous forces generate a vertically parallel shift in the total cost curve, the equilibrium price and quantity will not change, and thus this cost shifter will not help with identification. Only those demand shifters that cause a rotation in the demand curve will help identify the model.

Researchers who use the structural model have been willing to assume forms of demand and cost curves that aid in identification. Thus factor prices and incomes have been assumed to enter the cost and demand curves respectively in an interactive form with output. Despite this, the structural form has a tendency to produce results that are not economically meaningful. When this occurs, it is natural to assume that the researcher has not been able to discover the correct functional form for the revenue and total cost curves. However, there is another possibility: that the structural model itself is incorrect for the industry being investigated.

II. Limitations of the Structural Mode

The structural model of oligopoly is simple, spare, and elegant. The most powerful models in economics – the models of perfect competition and monopoly – are similarly minimal. However, the latter have been shown to be robust to minor

alterations in the structure of the model, while Hyde and Perloff show that a similar claim apparently can not be made for the structural model.

There is an appealing logic to the idea that the oligopoly outcome is intermediate between the competitive and monopoly result. The observer can give a knowing nod to the idea that the marginal revenue curve on which a member of an oligopoly acts will be between the two extremes. Unfortunately, this intuitive justification for the structural model, when examined more closely, is less compelling than it appears at first. The reader will remember that the way that the structural model starts with ignorance about firm costs and industry demands. It is only through the assumption that each observation represents an equilibrium and that the critical proportion between marginal revenue and price is the same in each of the observations that the firm cost and industry demand curves are able to be estimated.

It is, of course, not unusual to assume that each observation in an econometric study represents an equilibrium point. But in time series studies, as the period of adjustment becomes more compressed (for example if monthly or weekly data are used) it seems only sensible that the researcher deal with the question of how rapidly it is reasonable to assume that prices and quantities change in response to underlying cost and demand factors. A logical way to begin such a consideration would be the hypothesized mechanism in the model by which an industry out of equilibrium is presumed to return to equilibrium. If it is implausible for that process to come to equilibrium in a period as short as the time series observations cover, then doubt is cast on the assumption.

But what is the mechanism? It can not be simple profit maximization, since there is no way to reconcile individual maximizing decisions with an intermediate perceived marginal revenue curve, much less one that is based on a time-invariant proportion of the distance between price and industry marginal revenue. It is only through the assumption that industry pricing can be understood as the result of a game, like a one-shot symmetric Cournot quantity game, and that the rules of the game and the number of participants in the game do not change over the time of the study, that the postulate of the constant proportion can be justified. Unfortunately, the one-shot games that underlie the justification for constant λ should not properly be used to describe industry dynamics. That is, a game theoretic justification will not help give insight into the dynamics of λ.

In light of the silence of the structural model on the assumption of dynamics that underpin the assumption that observations are generated by an equilibrium in which there is, at every observation, a constant λ, it is worth considering some reasons for doubting the inherent logic of the assumption.

A. ADAPTATIONS WITHIN THE COST CURVE

There is a fundamental mismatch between the timing of pricing and investment decisions seen in the observable world and those assumed by the structural model. By estimating a cost curve as a function of output and factor prices,

$C = C(Q, W, \beta)$ where W is a vector of factor prices and β is a vector of parameters to be estimated, the structural model implicitly assumes that a firm facing a change in wage rates or interest rates will immediately lay off or hire labor, or build plants or sell off plants to adapt to the new optimum condition. While industries differ in the speed of adjustment, it is reasonable to assume that in the majority, it will take several years for a firm to adapt to new underlying conditions, rather than the months or weeks assumed by the structural model. The daily newspaper is filled with stories of corporate restructurings in which thousands of people are to be laid off over a period of several years in response to changes in a firm's perceived opportunity set. New physical investment in many industries can take a decade between proposal and the first production from the new plant. Employers report that finding and fully training a skilled labor force for a plant may take many years.

In view of the inevitable lags between changes in factor prices or market demand and actual movements to the new optimum on the expansion path for each firm, it seems unlikely that estimations based on the assumption that the market adapts within a week or a month will reflect reality.

B. The diversity of oligopoly pricing

Oligopoly pricing – especially the short run dynamics that most strongly influence time series data – is far more complex than is generally assumed by the game theoretic justification of the monopoly model. While it is reasonable to assume that the equilibrium oligopoly outcome is intermediate between monopoly and competition, it is quite implausible to believe that over time the result will be based on an same proportion over the years or decades from which the data are drawn. Consider, for example, the following reasons to doubt the premise of a constant λ.

- The structural model, like all models of industries producing homogeneous products into a single market, assumes that market power is held by the industry rather than by individual firms in the industry. As a contemporary example, consider the computer software industry, where the concern is not that the industry as a whole has market power, but that a single firm – Microsoft – has the ability to determine market conditions to buyers and other sellers. If the industry is organized into strategic groups (Caves and Porter, 1977) or if there is sufficient spatial or product differentiation to permit non-uniform pricing (Shepherd, 1972) then treating pricing as the result of the combined output decisions of all firms, rather than the private decision of an individual firm, will clearly lead to a misspecified model. Under these conditions, estimating a single total revenue function for the entire industry and assuming that a firm's only way to affect its market is to change its output, seriously misunderstands the nature of market power.

- If it is possible for an industry to build inventories as a buffer against uncertain demands, then there is a break in the direct connection assumed by the structural model between quantity as determining price and quantity of sales as determining costs. One reason for using inventories is to retain skilled labor in recessionary periods and to avoid the need to hire labor in periods of economic boom. There is, in fact, considerable evidence that oligopolies use inventories to break the connection between production and marketing (Kahn, 1992). But this means that an increase in sales in a week or month will not be reflected in production, leading to a mismeasurement of the cost curve, and with it, the entire structural model.
- Oligopoly prices are notably less volatile than those of competitive industries (Carlton, 1986). One explanation that has been offered for this fact is that industries use various devices for coordinating their prices (Scherer and Ross, 1990). Basing point pricing is the most obvious example of such a coordinating device, but there are many others: the use of rules of thumb and focal points and advanced announcements of price changes are only a few. The nature of these coordinating tools is that price changes will be made only infrequently, when market conditions clearly call for a price change. Since price changes are made infrequently, these changes will of course be non-incremental. But if an industry uses a coordinating device to make large discontinuous price changes, one can not expect that every week or month, the industry price will be determined by using a constant proportion of the difference between industry marginal revenue and market demand. On average, over many years, it may be the case that the oligopoly result will be between that of a monopoly and a competitive industry, but there is no reason to expect that an industry maintains the same distance between the two extremes at all times. By forcing λ to be constant, the model forces the estimated demand and cost curves to reflect factors that do not properly belong in them – thus perhaps leading to the sort of anomalies seen by Hyde and Perloff.
- Some industries have been analyzed by limit pricing models or models of predatory behavior (Geroski *et al.*, 1990). Neither motive is accounted for in the structural model. To the extent that the time series for an industry includes periods during which these motives determined behavior, the results will clearly be biased since prices will be set by external conditions not accounted for in either the current demand or cost curves.

The structural model follows a modern tradition of downplaying the difference between long and short run profit maximizing decisions (Baumol *et al.*, 1988.) To the extent that the distinction is introduced, it is done through introducing a fixed factor of production in the cost function. But this neglects the importance of long run considerations on the demand side. Promotional pricing has long been recognized as normal business conduct, despite the fact that it implies the acceptance of short run losses. Similarly, in times of excess demand, firms have been known to ration

products to their distributors rather than raise prices to market levels in the hope of maintaining customer goodwill and thus retaining sales after the period of shortage has passed. This practice has been rationalized by transactions costs (Williamson, 1989). Firms have been known to sell goods below short run average variable cost in times of slack demand rather than shut down the firm as profit maximization would apparently require in order to retain the continued patronage of a customer base. In all of these cases, firms will be violating the short run profit maximizing rules that lie behind the justification for the structural model.[2] To the extent that data are generated by such contradictory behavior, the structural model will be misspecified.

III. Conclusion

The structural model should take its place as one more tool for understanding oligopoly behavior, but that it should not be seen as the only logical tool for understanding that behavior, or even a necessarily reliable one. The model is incorrectly perceived as being able to deal with all forms of oligopoly pricing, from competition to monopoly and to do so without the need to use accounting estimates of marginal cost which are widely recognized as being flawed.

The structural model is an attempt to discover oligopoly behavior in economic data by imposing a rigid model. The core of this model is the assumption that each observation is generated by production and pricing decisions that constructs each firm's perceived marginal revenue by using a constant proportion of the difference between industry marginal revenue and price. By forcing the data to maintain this proportion week after week, month after month, the data are molded into a form that may look little like the truth. There is little reason to believe that oligopoly markets would retain this constant proportion over years and decades, and considerable reason to believe that they would not: limit pricing, the use of coordinating devices like pre-announcement of price changes, inventory holding, and most particularly product differentiation or other sources of market power exercised by individual firms are all practices that would lead an industry to base prices on variable differences industry marginal revenue and price.

It is ironic that the fundamental rationale for the structural model – the attempt to avoid the use of accounting measures of marginal cost when measuring market power – appeared at the same time that the courts have shown increased willingness in using accounting proxies to marginal cost in the assessment of oligopoly behavior. The Areeda-Turner rule, which accepts average variable cost as a reasonable approximation to marginal cost, has found wide acceptance in the economic profession. It is surely true that costs are endogenous to market structure, if only for reasons of X-inefficiency, and that they should thus be used with caution. But industrial organization is not the only field that is troubled by data that is not as

[2] These arguments have been developed with reference to cost standards for defining predation. See (Scherer, 1993).

accurate as would be desirable. In cost-benefit analysis, shadow prices are widely used where market data are seen to be unreliable, and similar adjustments could be made in cases where researchers have reason to believe that they are unreliable. Rather than scrap the use of accounting costs, it would be far more accurate to accept them as first approximiations and make reasonable adjustments, for example, by using international comparisons to eliminate the influence of X-inefficiencies.

Oligopoly pricing is indeed complex. There are many ways by with market power can be captured and be exercised. To assume that industries have only a single means for controlling market behavior misunderstands the observable world. There may be many industries which can be described by the structural model, but this observer believes that research effort would be better spent on the more mundane tasks of measuring empirical regularities associated with market power and producing reduced form results. This is an honored occupation in other social sciences, and even in medical science, and should be accepted in industrial economics as well. It is true that such exercises are not as satisfying because they can not be hung on a simple model, but they are less likely to be incorrect.

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